CHEAT SHEET UTS · WEEKS 1–7

MK 5 · Decision Making & Negotiation

Don't mix these up

- Availability = easy to recall; representativeness = looks typical; anchoring = stuck on a first number.
- Risk = known probabilities; uncertainty = unknown. MUST = screen; WANT = weight.
- KT: SA (priority: timing, trend, impact) · PA (past, IS/IS NOT) · DA (choose) · PPA (future: preventive vs contingent).
- AFT picks from what exists; VFT starts from values. Fundamental = why; means = how.
- Decision tree: roll back from the right, EMV = Σ p × payoff; EVPI = EV with perfect info − best EMV.
- AHP: 1–9 scale, reciprocal, CR ≤ 0.10, beware rank reversal. SMART: 0–100 values × swing weights.

W1 Introduction to Decision Making: Heuristics & Biases

What the class games show · Intuition and heuristics · What decision making is, and its levels · Two thinking systems at a glance · Heuristic → bias → antidote · Decision levels and approaches to studying decisions

- Bounded rationality, heuristics, biases.
- Availability, representativeness, anchoring.
- Risk ≠ uncertainty.
- Descriptive, normative, prescriptive.
- Strategic decisions are broad and irreversible.

W2 The Rational Framework; Intuition vs Deliberation

Decision-making styles · The six-step decision model · Reading: Intuition vs Deliberation · The three styles in one chart · MUSTs then WANTs: scoring the IT vendors · Two strategies to improve decisions

- Reflexive, reflective, consistent.
- A problem is a gap.
- Six-step model.
- MUST vs WANT.
- Certainty, risk, uncertainty.
- Shift to Type 2 or nudge Type 1.

FORMULAS

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ERP A 745 vs Upgrade 685
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W3 Kepner-Tregoe: SA, PA, DA, PPA

The four KT processes · Situation Appraisal · Problem Analysis: IS / IS NOT and root cause · Decision Analysis and Potential Problem Analysis · KT as a timeline · SA priority scores: Exxon Valdez · IS / IS NOT: Oh, Nuts!!! in one table · Decision Analysis: MUSTs, WANTs, then risk

- SA prioritises by timing, trend, impact.
- PA uses IS/IS NOT.
- DA: MUST screens, WANT weighs, then risk.
- PPA: preventive, contingent, triggers.
- Solve root causes.

W4 Alternative-Focused vs Value-Focused Thinking

Smart Choices and the role of objectives · AFT vs VFT · Objective hierarchies, means-ends networks, attributes · AFT vs VFT: where alternatives come from · Attributes: natural, proxy and constructed · Nuka Raya in numbers and a hierarchy

- AFT chooses; VFT creates.
- Reframe problems as opportunities.
- Fundamental vs means.
- Hierarchies vs networks.
- Natural, proxy, constructed attributes.
- The best option is often not on the first list.

W5 SMART: Simple Multi-Attribute Rating Technique

The SMART steps · Class example: Bandung office location · Value tree and weights for the office · Bandung in charts: where the scores come from · BUMDes: efficient frontier and RM-weight sensitivity

- Value tree, 0–100, swing weights, aggregate, sensitivity.
- Normalise weights.
- Separate cost; efficient frontier.
- Find weight crossovers.
- Compare cost per benefit point with its value.

FORMULAS

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Nilai agregat = Σ (bobot ternormalisasi × nilai atribut)
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W6 Decision Trees

Elements and construction · Worked example · Risk, utility and limitations · The example tree, redrawn · Simulator: EMV, indifference probability and EVPI · Utility: why people reject the highest EMV

- Decision, chance, payoff nodes.
- Roll back: EMV = Σ p × payoff.
- EVPI.
- Indifference probabilities.
- Expected utility for risk aversion.

FORMULAS

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EMV = Σ (probabilitas × payoff)
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EVPI = 6,4 - 4,4 = 2,0
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```
EU big = 0.60 · EU small = 0.73
```

W7 Analytic Hierarchy Process (AHP)

Hierarchy and Saaty's scale · Computing weights and consistency · AHP vs SMART, and criticisms · The supplier hierarchy · AHP calculator and global synthesis · Answer: the BUMDes exercise with AHP

- Goal → criteria → alternatives.
- 1–9 scale, reciprocal.
- Normalise columns, average rows.
- CR ≤ 0.10.
- Global = Σ weight × local.
- Beware rank reversal.

FORMULAS

```
CI = (λmax - n) ÷ (n - 1) = 0,039 ÷ 2 = 0,019    CR = CI ÷ RI = 0,019 ÷ 0,58 = 0,033
```

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λmax ≈ 4.008, CI 0.003, CR ≈ 0.003
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